

Experiment 20

Question

Perform Sharpening of Image using Laplacian Mask with Negative Center Coefficient.

Laplacian Mask

```
0 1 0
1 -4 1
0 1 0
```

Aim

To sharpen an image using the Laplacian mask with a negative center coefficient

Procedure

1. Import the OpenCV and NumPy libraries.
2. Read the input image (dog.jpg).
3. Create the Laplacian mask with **-4** as the center coefficient.
4. Apply the mask using cv2.filter2D().
5. Generate the sharpened image.
6. Display the original and sharpened images.
7. Save the output image.

Code

```
import cv2
import numpy as np

# Read image
img = cv2.imread("dog.jpg")

# Laplacian mask
kernel = np.array([[0,1,0],[1,-4,1],[0,1,0]])

# Apply filter
sharp = cv2.filter2D(img, -1, kernel)
```

```
# Display images
cv2.imshow("Original", img)
cv2.imshow("Sharpened", sharp)

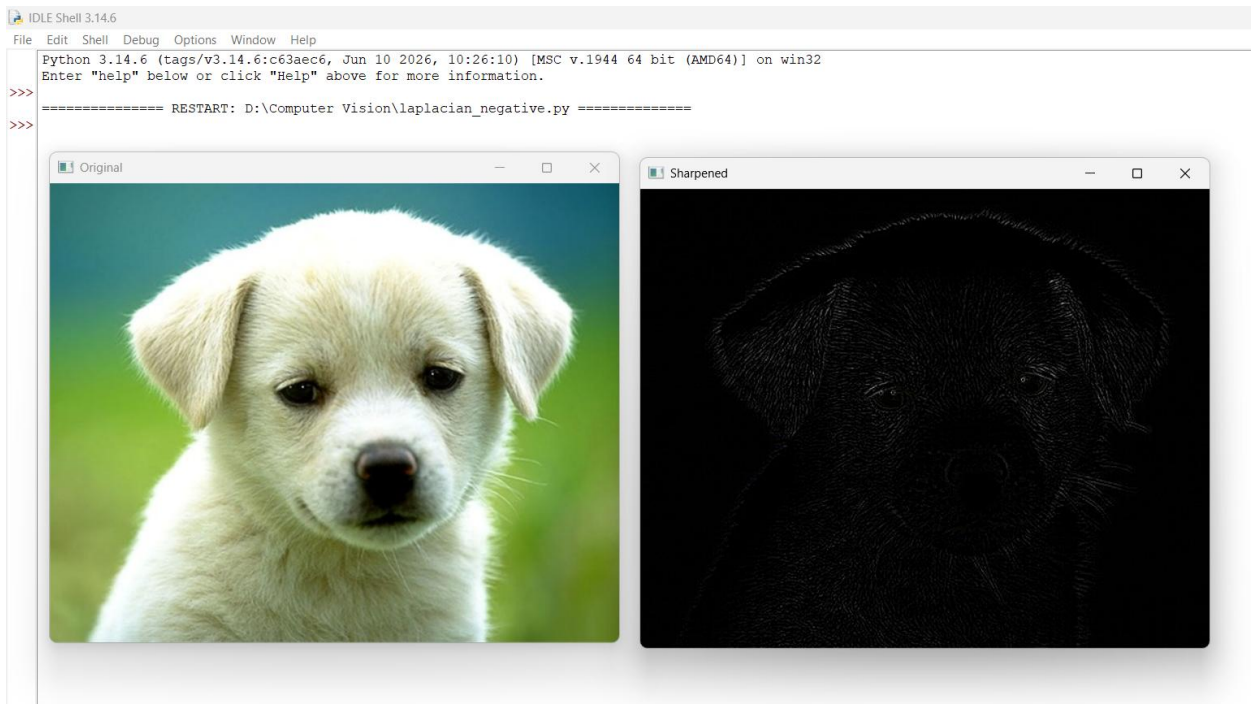
# Save image
cv2.imwrite("laplacian_negative.jpg", sharp)
```

Input

Input Image: dog.jpg

Output

- Sharpened Image



Result

The image was successfully sharpened using the Laplacian mask with a negative center coefficient.